EXPERIMENT – 5

AIM:

To create SQL views for department salary summary and employee hierarchy, test the updatability of views, and implement a recursive CTE to display reporting chains.

SOURCE CODE:

```
CREATE TABLE Department (
   Dept_ID INT PRIMARY KEY,
   Dept_Name VARCHAR(50),
   Location VARCHAR(50)
);

CREATE TABLE Project (
   Project_ID INT PRIMARY KEY,
   Project_Name VARCHAR(100),
   Budget DECIMAL(12,2)
);

CREATE TABLE Employee (
   Emp_ID INT PRIMARY KEY,
   Emp_Name VARCHAR(50),
   Salary DECIMAL(10,2),
   Job_Role VARCHAR(50),
   Dept_ID INT,
   Project_ID INT,
   Manager_ID INT,
   FOREIGN KEY (Dept_ID) REFERENCES Department(Dept_ID),
   FOREIGN KEY (Project_ID) REFERENCES Project(Project_ID)
);

INSERT INTO Department VALUES
(1, 'Human Resources', 'Delhi'),
(2, 'Information Technology', 'Chandigarh'),
(3, 'Finance', 'Mumbai'),
```

```
(4, 'Marketing', 'Bangalore'),
(5, 'Research and Development', 'Hyderabad');

INSERT INTO Project VALUES
(101, 'Website Development', 500000),
(102, 'Mobile Application', 750000),
(103, 'Cyber Security System', 900000),
(104, 'Data Analytics Platform', 850000),
(105, 'AI Chatbot', 1200000),
(106, 'Cloud Migration', 1000000),
(107, 'Payroll Management', 400000),
(108, 'Market Research System', 600000);

INSERT INTO Employee VALUES
(1, 'Aarav Sharma', 45000, 'HR Executive', 1, 107, NULL),
(2, 'Ananya Gupta', 52000, 'HR Manager', 1, 107, NULL),
(3, 'Riya Verma', 48000, 'Recruiter', 1, 107, 2),
(4, 'Karan Singh', 55000, 'HR Analyst', 1, 107, 2),
(5, 'Priya Mehta', 60000, 'HR Manager', 1, 107, NULL),
(6, 'Rahul Kumar', 75000, 'Software Engineer', 2, 101, 10),
(7, 'Neha Sharma', 82000, 'Software Engineer', 2, 102, 10),
(8, 'Aditya Singh', 95000, 'Security Engineer', 2, 103, 10),
(9, 'Simran Kaur', 88000, 'Data Analyst', 2, 104, 10),
(10, 'Vivek Gupta', 105000, 'Cloud Engineer', 2, 106, NULL),
(11, 'Ishita Jain', 78000, 'Software Developer', 2, 101, 10),
(12, 'Rohan Verma', 92000, 'AI Engineer', 2, 105, 10),
(13, 'Aman Yadav', 65000, 'Financial Analyst', 3, 107, 15),
(14, 'Sneha Kapoor', 72000, 'Accountant', 3, 107, 15),
(15, 'Mohit Agarwal', 85000, 'Finance Manager', 3, 107, NULL),
(16, 'Pooja Sharma', 68000, 'Financial Analyst', 3, 107, 15),
(17, 'Nikhil Jain', 90000, 'Senior Accountant', 3, 107, 15),
(18, 'Kriti Verma', 62000, 'Accountant', 3, 107, 15),
(19, 'Arjun Malhotra', 58000, 'Marketing Executive', 4, 108, 20),
(20, 'Tanya Singh', 70000, 'Marketing Manager', 4, 108, NULL),
(21, 'Yash Gupta', 62000, 'Marketing Analyst', 4, 108, 20),
(22, 'Muskan Sharma', 75000, 'Brand Manager', 4, 108, 20),  (23,
'Dev Kumar', 66000, 'Sales Executive', 4, 108, 20),
```

```
(24, 'Naina Kapoor', 80000, 'Marketing Analyst', 4, 108, 20),
(25, 'Kabir Sharma', 95000, 'Research Scientist', 5, 105, 28),
(26, 'Meera Gupta', 88000, 'ML Engineer', 5, 105, 28),
(27, 'Varun Singh', 92000, 'Research Analyst', 5, 104, 28),
(28, 'Aditi Verma', 110000, 'AI Engineer', 5, 105, NULL),
(29, 'Harsh Jain', 98000, 'Data Scientist', 5, 104, 28),
(30, 'Isha Kapoor', 85000, 'Research Engineer', 5, 103, 28);

CREATE VIEW Department_Salary_Summary AS
SELECT
   d.Dept_ID,
   d.Dept_Name,
   COUNT(e.Emp_ID) AS Employee_Count,
   AVG(e.Salary) AS Average_Salary,
SUM(e.Salary) AS Total_Salary,
   MAX(e.Salary) AS Highest_Salary,
   MIN(e.Salary) AS Lowest_Salary
FROM Department d
LEFT JOIN Employee e
ON d.Dept_ID = e.Dept_ID
GROUP BY d.Dept_ID, d.Dept_Name;

SELECT * FROM Department_Salary_Summary;

CREATE VIEW Employee_Hierarchy AS
SELECT
   e.Emp_ID,
   e.Emp_Name,
   e.Job_Role,
   e.Dept_ID,
   e.Manager_ID,
   m.Emp_Name AS Manager_Name
FROM Employee e
LEFT JOIN Employee m
ON e.Manager_ID = m.Emp_ID;

SELECT * FROM Employee_Hierarchy;
```

```
CREATE VIEW Employee_Basic_View AS
SELECT
   Emp_ID,
   Emp_Name,
   Salary,
   Dept_ID
FROM Employee;

UPDATE Employee_Basic_View
SET Salary = Salary + 1000
WHERE Emp_ID = 1;

SELECT * FROM Employee_Basic_View
WHERE Emp_ID = 1;

WITH RECURSIVE Reporting_Chain AS
(
   SELECT
      Emp_ID,
      Emp_Name,
      Manager_ID,
      0 AS Level
   FROM Employee
   WHERE Manager_ID IS NULL

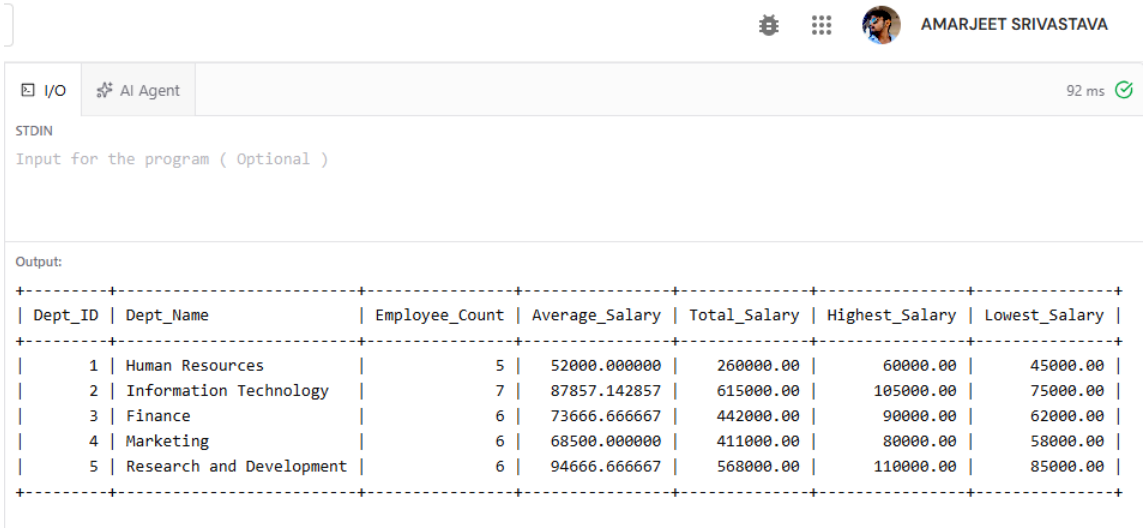
   UNION ALL

   SELECT
      e.Emp_ID,
      e.Emp_Name,
      e.Manager_ID,
      rc.Level + 1
   FROM Employee e
   INNER JOIN Reporting_Chain rc
   ON e.Manager_ID = rc.Emp_ID
)
```

```
SELECT
   Emp_ID,
   Emp_Name,
   Manager_ID,
   Level
FROM Reporting_Chain
ORDER BY Level, Emp_ID;
```

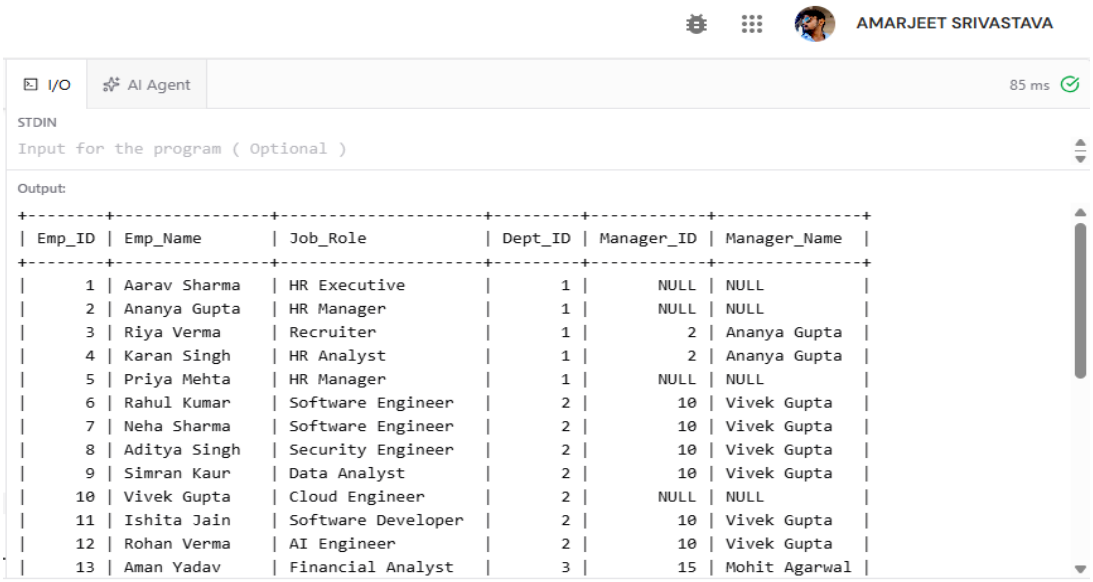
OUTPUT :

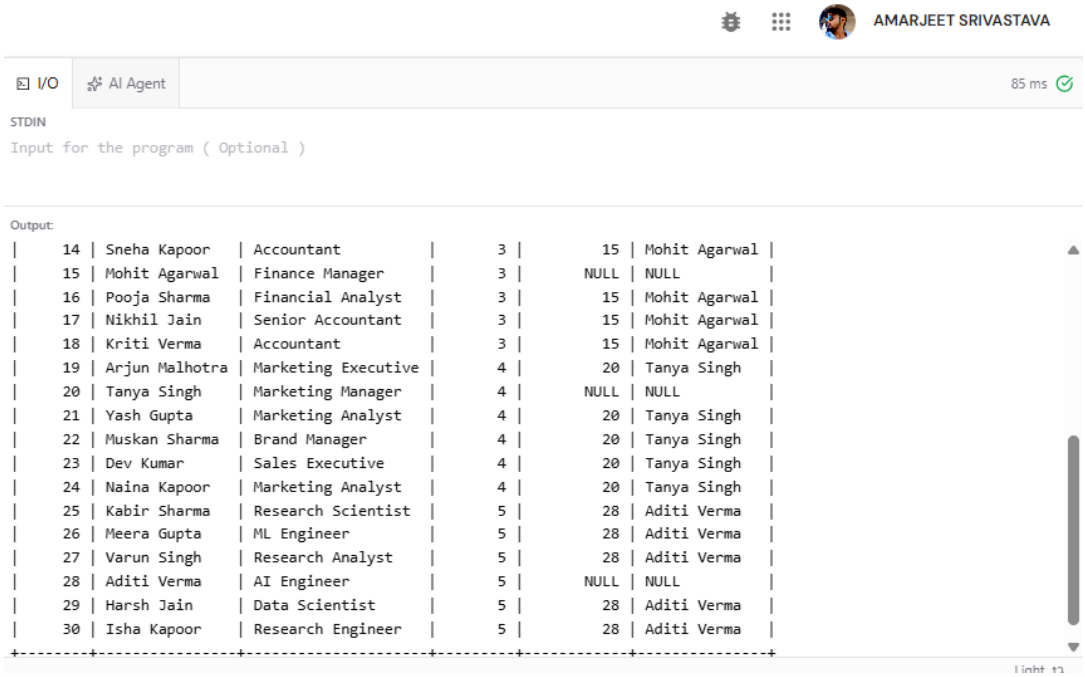
Department_Salary_Summary



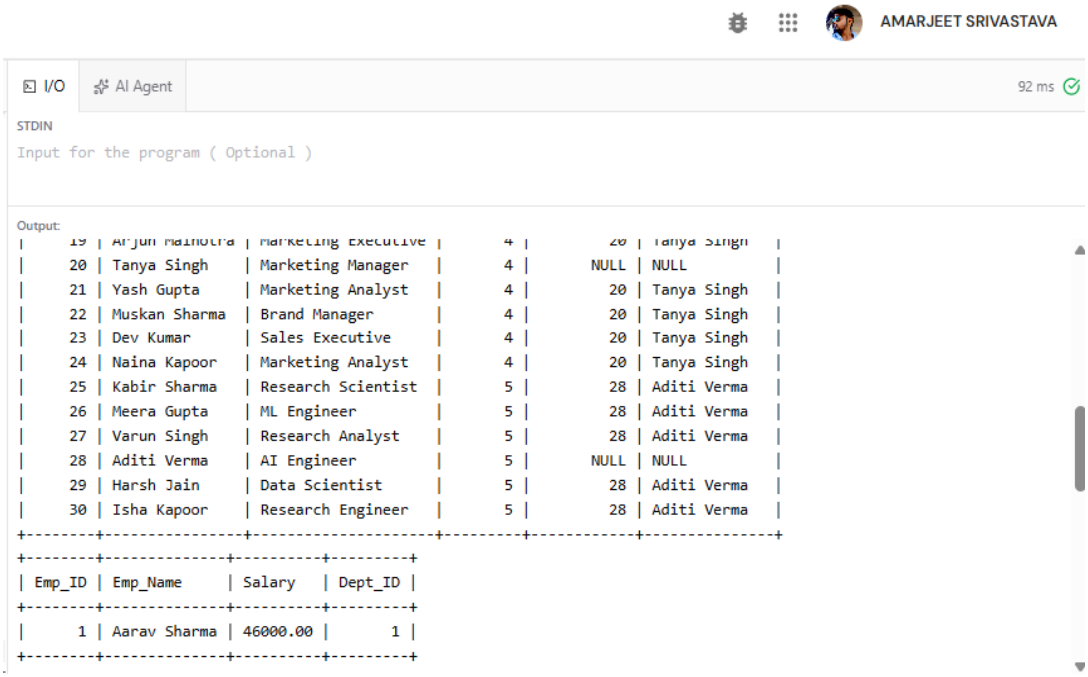
Dept_ID	Dept_Name	Employee_Count	Average_Salary	Total_Salary	Highest_Salary	Lowest_Salary
1	Human Resources	5	52000.000000	260000.00	60000.00	45000.00
2	Information Technology	7	87857.142857	615000.00	105000.00	75000.00
3	Finance	6	73666.666667	442000.00	90000.00	62000.00
4	Marketing	6	68500.000000	411000.00	80000.00	58000.00
5	Research and Development	6	94666.666667	568000.00	110000.00	85000.00

Employee_Hierarchy





Employee_Basic_View


AMARJEET SRIVASTAVA
I/O
AI Agent
92 ms
STDIN
Input for the program (Optional)
Output:
19	Arjun Malhotra	Marketing Executive	4	20	Tanya Singh
20	Tanya Singh	Marketing Manager	4	NULL	NULL
21	Yash Gupta	Marketing Analyst	4	20	Tanya Singh
22	Muskan Sharma	Brand Manager	4	20	Tanya Singh
23	Dev Kumar	Sales Executive	4	20	Tanya Singh
24	Naina Kapoor	Marketing Analyst	4	20	Tanya Singh
25	Kabir Sharma	Research Scientist	5	28	Aditi Verma
26	Meera Gupta	ML Engineer	5	28	Aditi Verma
27	Varun Singh	Research Analyst	5	28	Aditi Verma
28	Aditi Verma	AI Engineer	5	NULL	NULL
29	Harsh Jain	Data Scientist	5	28	Aditi Verma
30	Isha Kapoor	Research Engineer	5	28	Aditi Verma
Emp_ID	Emp_Name	Salary	Dept_ID
1	Aarav Sharma	46000.00	1

Reporting_Chain

